

Course Code	21CSE583T	Course Name	NATURAL LANGUAGE PROCESSING AND ITS APPLICATIONS	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
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Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Computing Technologies	Data Book / Codes/Standards			NA

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	Familiarize the basic concepts of NLP
CLR-2:	Inculcate the knowledge of preprocessing techniques of NLP
CLR-3:	Build probabilistic language models
CLR-4:	Brief about syntax and semantic techniques
CLR-5:	Design the application models of NLP

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	Comprehend the ambiguity, challenges, and applications of NLP	3	-	-
CO-2:	Demonstrate NLP techniques such as Part-of-Speech tagging, morphology, and text processing.	3	-	-
CO-3:	Create applications using probabilistic models.	3	-	-
CO-4:	Develop applications utilizing probabilistic models.	3	-	-
CO-5:	Develop NLP applications using machine learning and deep learning models.	3	-	-

Module-1 – NLP Overview	9 hours
History of NLP, Applications of NLP: Information Retrieval, Information Extraction, Question Answering, Sentiment Analysis Optical Character Recognition, Text Categorization Word Prediction, Speech Recognition Machine Translation, Introduction to Text preprocessing, Tokenization, Lemmatization and stemming, Feature extraction NLP terminology, Components of NLP, Term Frequency (TF), Inverse Document Frequency (IDF), Modeling using TF-IDF, Spam Filtering	
Module-2 – NLP Techniques	9 hours
Parts of Speech: Tagsets for English, POS (Part of Speech) tagging, POS (Part of Speech) tagging, Named entity recognition (NER), Language Model Vectorization, Word Embedding, Bag of Words, Continuous Bag of Words, Word Cloud - Pre-trained models, Word2Vec, Discovering Semantic relationship using, Word2Vec, Glove, Elmo embedding, Topic Modeling, Latent Dirichlet Allocation (LDA), Applications of LDA, Tutorial4 :Implementation of POS Tagging, NER	
Module-3 – Web Scraping & Language Models	9 hours
Basic HTML tags, BeautifulSoup, Web Scraping on HTML page, scraping ecommerce site, retrieve data with API, Modelling using N-Gram Model, Skip Gram, N-Gram, Document Similarity, Deep Learning models, Sequence modeling, Why Recurrence is necessary, Backpropagation through time (BPTT), Understanding gates, RNN, LSTM Basics, Multilayer LSTM, Bi-Directional LSTM.	
Module-4 – Large Language Models	9 hours
Attention Mechanism, Large Language Models, Encoder – Decoder Architecture, Transformers, BERT, XLNet, BART, GPT, Claude Gemma and Gemini APIs. Large Language models for Indian Languages – MURIL BERT, Multilingual models, mBERT, mBART.	
Module-5 – Introduction to Generative AI	9 hours
Generative AI and Large Language Models, Building Sarcasm Detection, End to End Case study, Generative AI using Hugging Face API, RAG model demonstration, Building Co-Pilots for health Care and Fin Tech, Handling Hallucination and data security.	

Learning Resources	1.Mugan, J., <i>Natural language text processing with Python: Hands-on NLP in Python using NLTK, spaCy, gensim, and scikit-learn.</i>, 2017 2.In Loonycorn (Firm). <i>From 0 to 1: Machine learning, NLP & Python: cut to the chase.</i> &PacktPublishing, 2017	3.Tunstall L., Werra L., and Wolf T, <i>Natural Language Processing with Transformers.</i> [ISBN-13: 978-1098136796], (available on O'Reilly), 2022 4.Dan Jurafsky, D. and James H. Martin, J., <i>Speech and Language Processing Links to an external site.</i> (3rd ed), 2019 5.Raaijmakers S. 2022. <i>Deep Learning for Natural Language Processing Links to an external site.</i> O'Reilly. [ISBN-13: 978-1617295447].
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Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1.Dr. Mariappan Vaithilingam, Brahmarsive Inc, Bengaluru, India	1.Ms. B. G. Sudha, Great Learning	1.Dr. C. N. Subalalitha, SRMIST